

METU EEE

Thesis Defence Entry

Graduating students from EE have to fill in the following form to announce their defence after arranging a date and place.

Student ID	Program Type	E-mail	Status
1341130	Ph.D.	e134113@metu.edu.tr	Awaiting approval
First Name (all caps should not be used)		Last Name (all caps should not be used)	
ÇAĞLAR		AKMAN	
Primary Area			
Robotics			
Other related area (if any)			
Choose other research area (if any)			
Other related area (if any)			
Choose other research area (if any)			
Title (please do NOT enter it in all caps and please do NOT start it with a new line at the beginning)			
AI-Powered DECISION SUPPORT SYSTEM FOR WILDFIRE DISASTER RESPONSE AND MANAGEMENT (DISASTER-AWARE)			
Abstract (please do NOT start your abstract with a new line at the beginning)			
Operational wildfire decision support systems choose from closed catalogues of predefined alternatives, cannot revise their rule bases during an incident, and rarely expose the reasoning behind their recommendations. This thesis closes these gaps with DisasterAware, a distributed, concept-driven fuzzy decision support framework built on two strictly separated cores: a physics-based simulation core and a decision core that reads it only through observations. Local agents turn ten confidence-gated observation features into five decision concepts, a sparse Takagi-Sugeno rule base maps them to six intervention families, and a non-inferential coordination layer ranks regions and shares capacity. The central contribution is the open decision space. A staged adaptation loop tunes rule parameters, instantiates rules on uncovered situations, and, when the base has no answer, prompts a pretrained language model as a schema-constrained proposer. Every modification must clear deterministic verification gates and a satisficing forecast test before it acts, so the doctrine grows from five seed rules only as the incident demands; a graduated fail-safe and a no-harm rule protect life-safety orders, and every decision remains traceable to its concepts, rules, and confidences. The contributions are tested end to end. The simulation core reproduces the front propagation of four historical fires with 86 to 94 percent agreement. On this validated ground the full open configuration reduces mean burned area by 82 percent against no action, nearly halves the loss of the static			
Supervisor Title	Supervisor Name		
Prof. Dr.	Mehmet Kemal Leblebicioğlu		
Defence Date	Defence Time		
31 Aug 2026	15 00		
Room or online meeting link			
D114			
Chairperson Title	Chairperson Name		
Prof. Dr.	Mehmet Önder Efe		
Member Title	Member Name		
Prof. Dr.	Klaus Werner Schmidt		
Member Title	Member Name		
Assoc. Prof. Dr.	Ahmet Güneş		